

Experiment No. 04:

Experiment Title: Interfacing of ultrasonic sensor with Arduino.

Objectives

- Interface an HC-SR04 ultrasonic distance sensor with an Arduino Uno.
- Understand the principle of distance measurement using ultrasonic sound waves (time-of-flight method).
- Measure signal pulse durations using the pulseIn() function in embedded C/C++.
- Calculate the precise distance to an obstacle in centimeters and display the live data on the Serial Monitor.

Introduction

The HC-SR04 is an ultrasonic distance sensor widely used for obstacle detection and distance measurement. It operates on the sonar principle by emitting high-frequency sound waves (40 kHz) and measuring the time taken for the echo to bounce back from an object.

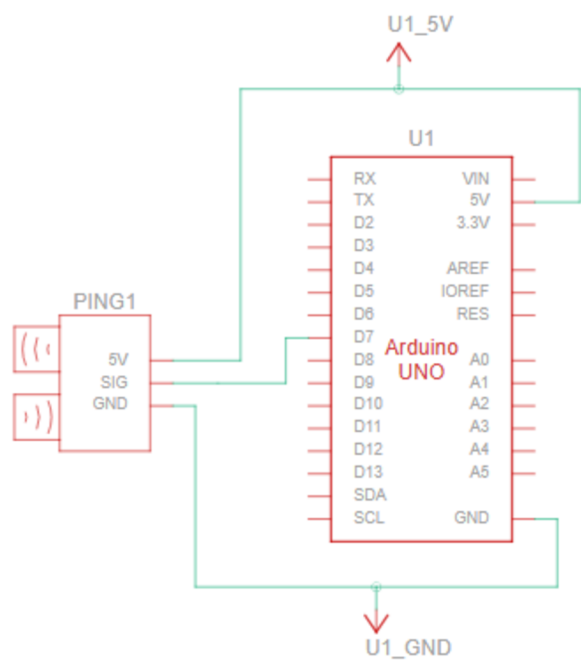
- Sensor Pins & Working Principle:
- VCC & GND: Power pins (operates at 5V DC).
- Trig (Trigger Pin): Receives a 10 μ s HIGH pulse from the Arduino to emit an 8-cycle sonic burst.
- Echo (Echo Pin): Outputs a HIGH pulse whose duration (μ s) is directly proportional to the distance traveled by the sound wave.
- Distance Calculation Formula:
- Speed of sound in air is approximately 340 m/s or 0.034 cm/ μ s. Since the wave travels to the object and back, the total distance is halved:

$$\text{Distance (cm)} = \frac{\text{Echo Time } (\mu\text{s}) \times 0.034}{2}$$

Apparatus Required

Component / Equipment	Quantity	Specification / Description
Arduino Uno R3	1	Microcontroller development board
HC-SR04 Ultrasonic Sensor	1	4-pin ultrasonic distance module
Breadboard	1	Solderless prototyping board
Jumper Wires	As needed	Male-to-Male / Male-to-Female jumper wires

Circuit Diagram



Working Procedure

1. Drag and place the Arduino Uno and HC-SR04 Ultrasonic Sensor onto the Tinkercad workspace.
2. Connect the VCC pin of the HC-SR04 to the 5V power rail on the Arduino.
3. Connect the GND pin of the HC-SR04 to the GND rail on the Arduino.
4. Connect the Trig pin of the HC-SR04 to Digital Pin 9 of the Arduino.
5. Connect the Echo pin of the HC-SR04 to Digital Pin 10 of the Arduino.
6. Open the Tinkercad code window, select C++ text mode, and write the program.
7. Click Start Simulation, open the Serial Monitor, and click on the HC-SR04 sensor in Tinkercad to move the target obstacle and observe real-time distance measurements.

Code

```
int Pin = 7;

void setup() {
    Serial.begin(9600);}

void loop() {
    long duration, cm;
    pinMode(Pin, OUTPUT);
    digitalWrite(Pin, LOW);
    delayMicroseconds(2);
    digitalWrite(Pin, HIGH);
    delayMicroseconds(5);
    digitalWrite(Pin, LOW);
    pinMode(Pin, INPUT);
    duration = pulseIn(Pin, HIGH);
    cm = (duration/29) / 2;
    Serial.print("Distance: ");
    Serial.print(cm);
    Serial.println("cm");
    if(cm >= 50){ Serial.println("\tFar from Sensor");}
    delay(1000);}
```

Discussion and Conclusion

- Discussion: The HC-SR04 sensor successfully interfaced with the Arduino Uno to measure distances accurately. Pulsing the trigPin HIGH for 10 microseconds initiated the ultrasonic burst. The duration returned by pulseIn(echoPin, HIGH) represented the round-trip travel time of the sound wave. Applying the formula $\text{Distance} = (\text{Duration} \times 0.034) / 2$ converted this time measurement into accurate centimeter values displayed live on the Serial Monitor.
- Final Verdict: The experiment was completed successfully. The Arduino accurately triggered the ultrasonic sensor, captured the echo signal timing, and computed object distance using the time-of-flight principle.